

Microplastic Polymer Analysis Report

Report ID	c9f73a64-3d05-4d2a-8181-c0e1dda7c82b
Generated On	19 May 2026 14:21
Status	Completed
Sample ID	FTIR-4272

Prediction Results

Detected Polymer	HDPE
Confidence Score	0.94%
Model Used	finetuned_model

FTIR Spectral Analysis

- 2915 cm^{-1} → CH_2 asymmetric stretching
- 1470 cm^{-1} → CH_2 bending

Confidence Interpretation

■■■■■■■■■■ 0.94%

Scientific References

- [1] Harper, C. Handbook of Plastics Technologies.
- [2] Thompson, R. Plastic debris in oceans.
- [3] Barnes, D. Accumulation of plastics.
- [4] Andrady, A. Environmental impacts of plastics.
- [5] Jung, M. FTIR analysis of HDPE.

AI Generated Summary

The analyzed sample was identified as HDPE with a confidence score of 0.94%. The FTIR spectral characteristics strongly support the presence of HDPE polymer chains. The detected functional groups and absorption bands are

consistent with known spectral signatures of HDPE. The analysis indicates a reliable classification suitable for environmental and material identification studies.